WORKED SOLUTION

STEP III 1997, Question 9

Mechanics • Statics with Elastic Springs

STEP III 1997, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q9

The key idea

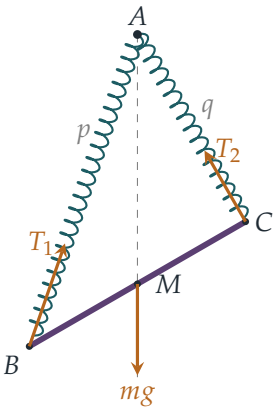
Only three forces act on the rod: the two spring tensions and its weight. When a body is in equilibrium under three non-parallel forces, their lines of action must pass through a single point. Both tensions point along lines through A, so the weight must pass through A too, which puts the midpoint of the rod directly below A. After that, horizontal resolution almost does the rest.

Solution

We model the rod as uniform and rigid, with mass m and midpoint M. The springs are light, obey Hooke's law with the same constant κ, and are both stretched. Let the springs AB and AC have natural lengths a and b and stretched lengths p and q, so the tensions are

$$T_1 = \kappa(p - a), \qquad T_2 = \kappa(q - b),$$

where both are positive, since the springs are stretched.



Step 1: M is directly below A

Take moments about A for the rod. The tension T_1 acts along BA and T_2 along CA, so both lines pass through A and neither has a moment about A. Hence the weight, acting vertically through M, must also have zero moment about A, so the vertical line through M passes through A. Thus M lies directly below A; let $h > 0$ be its depth below A.

Step 2: resolve horizontally

Set up axes with origin A, x horizontal (in the vertical plane of the rod) and y vertically upwards. Then $M = (0, -h)$, and since M is the midpoint of BC we can write

$$B = (-d,\ -h - e), \qquad C = (d,\ -h + e)$$

for some d, e. Here d is the horizontal half-width of the rod, and $d \neq 0$ exactly because the rod is not vertical.

The tension in AB acts on the rod at B in the direction from B to A, that is, along the vector $\overrightarrow{BA} = (d,\ h + e)$, which has length p. So this force is $\dfrac{T_1}{p}(d,\ h + e)$. Similarly, the tension at C is $\dfrac{T_2}{q}(-d,\ h - e)$.

Resolving horizontally,

$$\frac{T_1}{p}\, d - \frac{T_2}{q}\, d = 0,$$

and since $d \neq 0$,

$$\frac{T_1}{p} = \frac{T_2}{q}.$$

Ask yourself Where exactly did we use the fact that the rod is not vertical?

Only in cancelling d. If the rod hung vertically, both springs would lie along the same vertical line, horizontal resolution would read $0 = 0$, and the tensions could be in any ratio, subject only to the vertical balance. Such vertical equilibria really do occur, and in them the result is generally false.

Step 3: bring in Hooke's law

Substituting the tensions,

$$\frac{\kappa(p - a)}{p} = \frac{\kappa(q - b)}{q} \quad \Longrightarrow \quad 1 - \frac{a}{p} = 1 - \frac{b}{q} \quad \Longrightarrow \quad \frac{a}{p} = \frac{b}{q},$$

so

$$\frac{p}{q} = \frac{a}{b}.$$

The ratio of the stretched lengths equals the ratio of the natural lengths, as required. ■

For completeness, resolving vertically gives $\dfrac{T_1}{p}(h+e)+\dfrac{T_2}{q}(h-e)=mg$, that is, $2h\,\dfrac{T_1}{p}=mg$. We don't need this, but it fixes the common value of T/length.

Checks

- *Interpretation.* Since $p/q=a/b$, each spring is stretched by the same factor: $p=\lambda a$ and $q=\lambda b$ for some $\lambda>1$. So the two springs behave like one uniform spring scaled up in proportion, which is a pleasing way to remember the answer.
- *Numerical.* We found equilibria by minimising the potential energy $mgy_M+\frac{1}{2}\kappa(p-a)^2+\frac{1}{2}\kappa(q-b)^2$ numerically for random m,κ,a,b and rod lengths. For example, with $a/b=0.709972$, a tilted equilibrium gave $p/q=0.709972$, with M directly below A to within 10^{-7}. The same search also turned up vertical equilibria with $p/q\neq a/b$ (for example, $a/b=1.526$ but $p/q=1.192$), which confirms that the hypothesis is needed.

Common slips

- Assuming the tensions are equal, or that the springs make equal angles with the vertical. Neither is true in general; what matches is tension divided by length.
- Taking moments about M and getting tangled in angles. Moments about A are much cleaner, because both tensions pass through A.
- Never using the condition that the rod is not vertical. A complete answer says where it is needed.

Beyond STEP

The vector $\dfrac{T}{p}\overrightarrow{BA}$ we used is the natural way to write a spring force. A spring with zero natural length exerts a force exactly proportional to the displacement vector, and such "ideal" springs turn many equilibrium problems into linear algebra. More generally, an equilibrium is a stationary point of the potential energy, which is the idea behind our numerical check and the starting point of Lagrangian mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both statics and the energy viewpoint carefully.

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